

# PMx Weekly Digest: July 04, 2026

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## Paper 1: Integrated Population Pharmacokinetic Analysis of Intravitreal Aflibercept Drug

### Bibliographic Information

- **Title:** Integrated Population Pharmacokinetic Analysis of Intravitreal Aflibercept Drug Products
- **Authors:** Bihorel S, Chiu J, Eissing T, et al.
- **Journal:** Clinical Pharmacology & Therapeutics, 2025
- **DOI:** <https://doi.org/10.1002/cpt.70000>

**Paper Type:** POPK\_PKPD

### Quick Take

This integrated PopPK analysis of 16 clinical trials (n=2744) reveals that the high-dose aflibercept formulation (AFL-HD, 8 mg) has a 39.4% slower ocular clearance than the standard 2-mg product (AFL-2), an effect attributed to drug-product differences beyond dose alone. This supports extended dosing intervals (every 12–16 weeks) for nAMD and DME. Pharmacometricians should read this for its rigorous semi-mechanistic modeling approach and the surprising finding that formulation characteristics can influence ocular pharmacokinetics independently of dose.

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### Executive Summary

This paper presents an integrated population pharmacokinetic (PopPK) analysis of free and bound aflibercept in plasma from 2744 participants across 16 clinical trials, including the pivotal PULSAR and PHOTON studies. A semi-mechanistic model was developed to describe ocular release, systemic disposition, and target-mediated elimination of aflibercept after intravitreal (IVT), subcutaneous (SC), and intravenous (IV) administration. The model revealed that the high-dose formulation (AFL-HD, 8 mg) exhibits a 39.4% slower ocular clearance (QE) compared to the standard 2-mg product (AFL-2), an effect attributed to drug-product differences beyond dose alone. Covariate effects (weight, age, sex, albumin, disease, race, renal/hepatic function) were small and not clinically meaningful. The slower QE for AFL-HD extends ocular exposure, supporting extended dosing intervals (every 12–16 weeks) and reduced treatment burden for neovascular age-related macular degeneration (nAMD) and diabetic macular edema (DME).

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### Scientific Context & Motivation

Intravitreal anti-VEGF therapies are standard of care for nAMD and DME but require frequent injections every 4–8 weeks, creating a significant treatment burden. Higher molar doses may prolong therapeutic effect and extend dosing intervals. Aflibercept is available as a 2-mg product (AFL-2) and a new 8-mg product (AFL-HD) with a different formulation. Published PK data after IVT injections are limited. This study addresses the gap by quantifying the release of aflibercept

from the eye after IVT administration of both drug products and identifying factors influencing ocular pharmacokinetics.

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## ✂ Methodological Snapshot

The authors developed a semi-mechanistic population PK model for free and bound aflibercept using data from 16 clinical trials (n=2744) after IVT, IV, and SC administration. The model includes compartments for the study eye, fellow eye, central and peripheral compartments, and a platelet compartment for nonlinear elimination. Ocular clearance (QE) was estimated as a bidirectional linear process. Covariate effects were evaluated using stepwise forward inclusion and backward elimination. The model was validated using pcVPCs, bootstrap, and sensitivity analyses. The key finding was a 39.4% slower QE for AFL-HD versus AFL-2, attributed to drug-product differences beyond dose alone.

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## ☒ Structural Model Breakdown

The model is a semi-mechanistic multi-compartment model with the following structure: Study eye compartment (V1, fixed 4 mL) and fellow eye compartment (V5, fixed 4 mL) connected to the central compartment via bidirectional linear ocular clearance (QE). Central compartment for free aflibercept (V2) and bound aflibercept (V4) assumed equal volumes. Two peripheral compartments (V3, V8) for free aflibercept. First-order elimination from central compartment ( $K_{20}$ ). Nonlinear elimination pathway via platelet compartment (V7) with Michaelis-Menten binding ( $V_{max,p}$ ,  $K_{m,p}$ ) and first-order elimination from platelet ( $K_{70}$ ). SC absorption compartment (V6) with first-order absorption rate ( $K_{62}$ ). Bound aflibercept elimination from central compartment ( $K_{40}$ ). Allometric scaling on V2, V3, V4, V8,  $K_{20}$ ,  $K_{40}$ , QF1, QF2 with body weight exponents estimated. Covariate effects on QE: age (power coefficient -1.36), drug product (proportional effect 0.606 for AFL-HD).

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## Detailed Methodological Analysis

### Modeling Approach

Semi-mechanistic model: multi-compartment model for free aflibercept with nonlinear binding to VEGF (Michaelis-Menten), first-order elimination ( $K_{20}$ ), and additional nonlinear elimination (platelet uptake). Bound aflibercept elimination ( $K_{40}$ ). Ocular clearance (QE) for bidirectional transfer between eye and central compartment. SC absorption ( $K_{62}$ ). Allometric scaling on volumes and clearances. Fixed eye volume (4 mL).

### Data Sources

Data from 16 clinical trials: 76 healthy individuals, 1662 nAMD, 968 DME, 38 solid tumor/lymphoma. Routes of administration: IV (0.3–4 mg/kg), SC (0.025–2 mg/kg), IVT (0.05–4 mg for AFL-2, 8 mg for AFL-HD). Sparse sampling in most participants; dense sampling in a subset. Total of 31,326 samples analyzed (15,657 free, 62.9% BLQ; 15,669 bound, 19.6% BLQ).

### **Estimation Methods**

Nonlinear mixed-effects modeling using NONMEM 7.5.0 with FOCE interaction. Beal's M3 method for BLQ data. Bootstrap (200 samples) for parameter precision. Sensitivity analyses: excluding AFL-HD data, removing IIV on QE, substituting drug product effect with continuous dose function.

### **Model Evaluation**

Goodness-of-fit (GoF) plots: observed vs. predicted, conditional weighted residuals vs. time, individual predictions. Prediction-corrected VPCs (pcVPC) with 1000 replicates. Comparison of simulated vs. observed data under prototypical dosing regimens. Sensitivity analyses confirmed robustness of drug product effect on QE.

### **Covariate Analysis**

Covariates evaluated included body weight, age, sex, albumin, disease (nAMD vs. DME), racial classification, Japanese origin, renal function (CKD-EPI, eGFR), and hepatic function (AST, ALT, bilirubin). Selection used stepwise forward inclusion ( $\alpha = 0.01$ ) and backward elimination ( $\alpha = 0.001$ ). Body weight was included via allometric scaling. Age and drug product effects on QE were retained. Disease effect on QE was not significant after adjusting for age. Covariate effects were small and not clinically meaningful.

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### **Statistical Rigor Assessment**

The analysis uses appropriate methods: FOCE interaction for NLME, Beal's M3 for BLQ data, stepwise covariate selection with appropriate significance thresholds ( $\alpha = 0.01$  for forward,  $\alpha = 0.001$  for backward), and bootstrap for parameter precision. The large sample size ( $n=2744$ ) provides adequate power for covariate detection. Sensitivity analyses are thorough: excluding AFL-HD data, removing IIV on QE, substituting drug product effect with continuous dose function, and excluding high-shrinkage data. The moderate shrinkage on QE (34.4%) is acknowledged and shown not to influence the drug product effect. Missing data (BLQ) are handled appropriately via M3 method. The analysis is rigorous and meets high standards for regulatory submissions.

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### **Key Findings**

The primary finding is that AFL-HD (8 mg) has a 39.4% slower ocular clearance (QE) compared to AFL-2 (2 mg), an effect attributed to drug-product differences beyond dose alone. QE decreases with increasing age (power coefficient  $-1.36$ ). After adjusting for age, QE is comparable between nAMD and DME. Body weight has the largest effect on systemic exposure (up to 52% increase at 5th percentile weight), but all other covariates (age, sex, race, albumin, renal function, hepatic function) have small effects ( $<25\%$ ). Free aflibercept shows more than dose-proportional increase in AUC (10.5-fold for 4-fold dose increase) due to nonlinear elimination. The model predicts that median ocular concentration at the end of the 8-week interval for AFL-2 is reached approximately 14 weeks after AFL-HD dosing, compared to 9–9.5 weeks if QE were identical.

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## 🔗 Clinical & Regulatory Implications

The model supports the extended dosing intervals of AFL-HD (8 mg every 12–16 weeks) compared to AFL-2 (2 mg every 8 weeks) for nAMD and DME. No dose adjustments are warranted based on age, weight, sex, race, renal function, or hepatic function, as covariate effects on systemic exposure were small (<25%). The slower QE for AFL-HD translates to longer ocular exposure, which underpins the non-inferior visual gains observed in pivotal trials with fewer injections. Clinicians can expect that the median ocular concentration at the end of the 8-week interval for AFL-2 is reached approximately 14 weeks after AFL-HD dosing, supporting the extended regimen.

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## Strengths & Limitations

### Strengths

- Large integrated dataset (16 trials, n=2744) with diverse routes of administration.
- Semi-mechanistic model captures complex PK of free and bound aflibercept.
- Rigorous sensitivity analyses confirm robustness of drug product effect on QE.
- Thorough covariate analysis with appropriate statistical methods.
- Model validated using pcVPCs, bootstrap, and comparison to observed data.

### Limitations (Acknowledged by Authors)

- The mechanistic basis for the slower QE for AFL-HD is not fully understood and warrants further investigation.
- Ocular concentrations were predicted from plasma data, not directly measured.
- Fixed eye volume (4 mL) assumed.
- Moderate shrinkage on QE (34.4%) may affect individual parameter estimates.
- The model assumes bidirectional linear clearance for QE.

### Limitations (Expert Review)

- No direct ocular concentration measurements; predictions rely on model assumptions.
- The model does not account for potential differences in vitreous humor volume or injection technique.
- The platelet elimination pathway is hypothesized and not directly validated with platelet data.
- The impact of fellow eye treatment on PK is not fully explored.
- The model does not include pharmacodynamic endpoints (e.g., visual acuity, OCT) for a full PK/PD link.

### Generalizability

The findings are generalizable to patients with nAMD and DME treated with AFL-2 or AFL-HD. The model includes a wide range of patient demographics and comorbidities. However, generalizability to other retinal diseases (e.g., diabetic retinopathy, macular edema following retinal vein occlusion) or other anti-VEGF agents (e.g., ranibizumab, bevacizumab) is limited without further validation.

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## Key Equations

### Ocular Clearance Model

$$\frac{dA_{eye}}{dt} = -QE \cdot C_{eye} + QE \cdot C_{central}$$

Bidirectional ocular clearance (QE) describing transfer of free aflibercept between the eye and central compartment.

### Target-Mediated Elimination Model

$$\frac{dA_{bound}}{dt} = \frac{V_{max} \cdot C_{free}}{K_m + C_{free}} - K_{40} \cdot C_{bound}$$

Michaelis-Menten-type binding of free aflibercept to VEGF in the central compartment.

### Allometric Scaling Model

$$CL_i = CL_{pop} \cdot \left( \frac{WT_i}{70} \right)^{\theta_{CL}} \cdot e^{\eta_{CL}}$$

Allometric scaling of clearance and volume parameters with body weight.

### Age Effect on QE

$$QE_{AFL-2} = \theta_{QE} \cdot \left( \frac{Age}{70} \right)^{\theta_{age}}$$

Effect of age on ocular clearance (QE).

### Drug Product Effect on QE

$$QE_{AFL-HD} = \theta_{QE} \cdot \theta_{AFL-HD} \cdot \left( \frac{Age}{70} \right)^{\theta_{age}}$$

Drug product effect on ocular clearance.

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## Figures & Tables

- **Figure 1:** Semi-mechanistic model structure showing compartments for study eye (V1), fellow eye (V5), central compartment (V2/V4), peripheral compartments (V3, V8), and platelet compartment (V7), with bidirectional ocular clearance (QE) and first-order absorption from SC site ( $K_{62}$ ).
  - *Significance:* Provides the structural framework for the entire analysis, illustrating the complex interplay between ocular release, systemic distribution, and target-mediated elimination.
- **Figure 2:** Prediction-corrected VPCs for free and bound aflibercept stratified by route of administration and IVT dose.
  - *Significance:* Demonstrates adequate model predictive performance across diverse dosing conditions, including data above and below LLOQ.

- **Figure 3:** Comparison of observed and model-predicted median concentrations over time for free and bound aflibercept under prototypical dosing regimens.
  - *Significance:* Validates the model’s ability to capture the time course of systemic exposure for both drug products, supporting the extended dosing intervals for AFL-HD.
- **Figure 4:** Forest plot of covariate effects on AUC and  $C_{max}$  for free and bound aflibercept under 8q12 dosing in prototypical patients.
  - *Significance:* Quantifies the impact of key covariates (weight, age, sex, renal function, etc.) on systemic exposure, showing that only body weight has a moderate effect (>25%).
- **Table 1:** Table of final PK parameter estimates with 95% CIs, RSE, CV, and shrinkage.
  - *Significance:* Central repository of all model parameters, including the key finding of 39.4% slower QE for AFL-HD and the age effect on QE.
- **Table 2:** Summary of predicted PK parameters after first dose of AFL-2 (2 mg) and AFL-HD (8 mg) in dense PK substudies.
  - *Significance:* Provides clinically relevant exposure metrics (AUC,  $C_{max}$ ,  $C_{trough}$ ,  $T_{max}$ , accumulation ratio) for both drug products, showing more than dose-proportional increase for free aflibercept.

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### Code & Reproducibility Assessment

The analysis used NONMEM 7.5.0, PsN 4.6.0, SAS 9.4, and R 4.1.2. The mrgsolve R package was used for simulations. No public code repository is mentioned. Qualified researchers may request access to study documents via Vivli.

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### Future Directions

The mechanistic basis for the slower QE for AFL-HD warrants further investigation, potentially through physicochemical characterization of the formulation (viscosity, injection volume, excipient effects) and preclinical ocular PK studies. Validation of predicted ocular concentrations with direct measurements from aqueous humor or vitreous humor samples would strengthen the model. The model could be extended to other retinal vascular diseases (e.g., macular edema following retinal vein occlusion, diabetic retinopathy) and to other IVT anti-VEGF agents (e.g., faricimab, ranibizumab). The impact of age on QE should be explored in other IVT drug classes. Finally, the model could be used to optimize individualized dosing strategies based on patient characteristics.

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### Expert Commentary

This is a tour de force in pharmacometrics—integrating data from 16 trials, three routes of administration, and two analytes (free and bound) into a single semi-mechanistic model. The use of IV and SC data to inform systemic disposition and nonlinear elimination is a clever strategy to overcome the sparsity of IVT data (72% BLQ for free aflibercept). The finding that AFL-HD has a 39.4% slower QE than AFL-2, independent of dose, is both surprising and impactful. It suggests that formulation characteristics (e.g., viscosity, injection volume, or excipients) can influence ocular pharmacokinetics—a lesson for future IVT drug development. The sensitivity analyses confirming

this effect are rigorous. The covariate analysis is thorough, but the small effects observed are reassuring for clinical practice. The model's ability to predict ocular concentrations from plasma data is a methodological highlight. One limitation is the lack of direct ocular concentration measurements; the model assumes a fixed eye volume (4 mL) and bidirectional linear clearance. Future work should validate these predictions with aqueous humor data. Overall, this paper sets a high standard for PopPK analysis of IVT biologics.

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### **Bottom Line**

This integrated PopPK analysis of 16 clinical trials (n=2744) provides a robust semi-mechanistic model for free and bound aflibercept after IVT, IV, and SC administration. The key finding—a 39.4% slower ocular clearance (QE) for the high-dose formulation (AFL-HD) versus the standard 2-mg product (AFL-2)—is attributed to drug-product differences beyond dose alone. This supports extended dosing intervals (every 12–16 weeks) for AFL-HD in nAMD and DME, potentially reducing treatment burden. Covariate effects (weight, age, renal function) were small and not clinically meaningful, requiring no dose adjustments.

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### **☒ Figures**

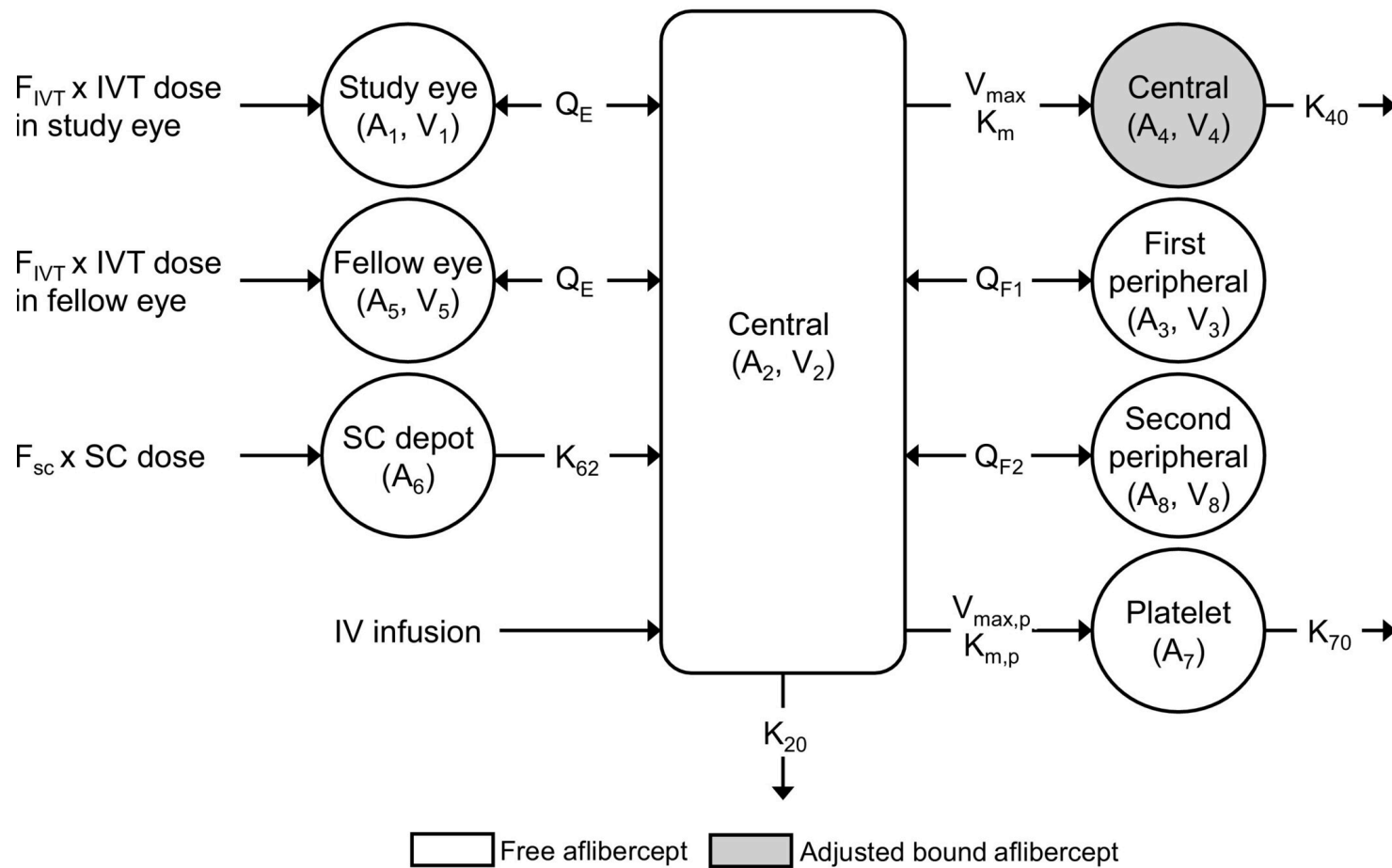


Figure 1: Structural representation of a population pharmacokinetic model following IVT, IV, and SC administration of aflibercept. Note that the central compartment for fr



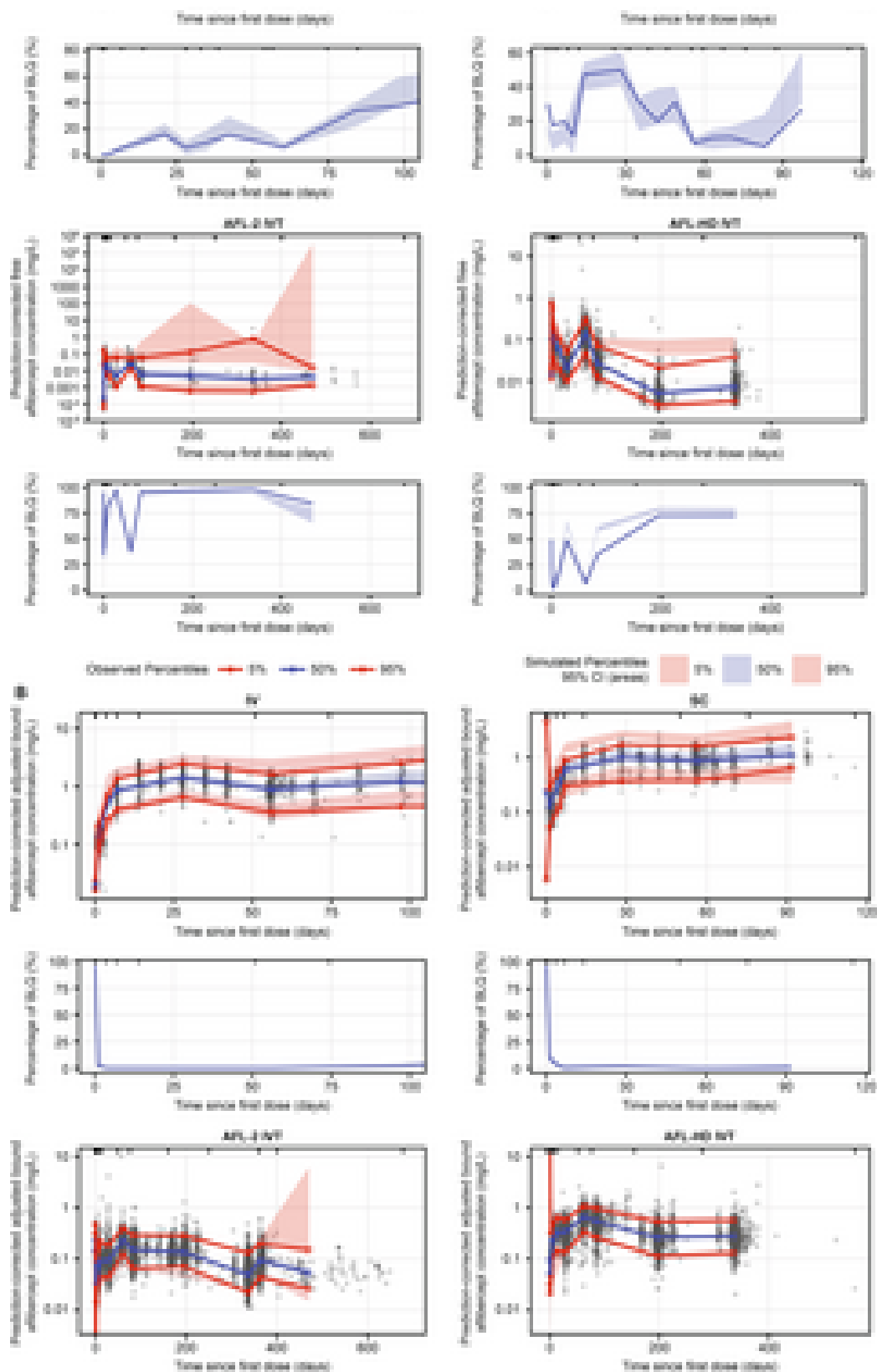


Figure 2: Prediction-corrected visual predictive checks for (a) free aflibercept and (b) adjusted bound aflibercept concentrations in plasma BLQ, below the lower limit of

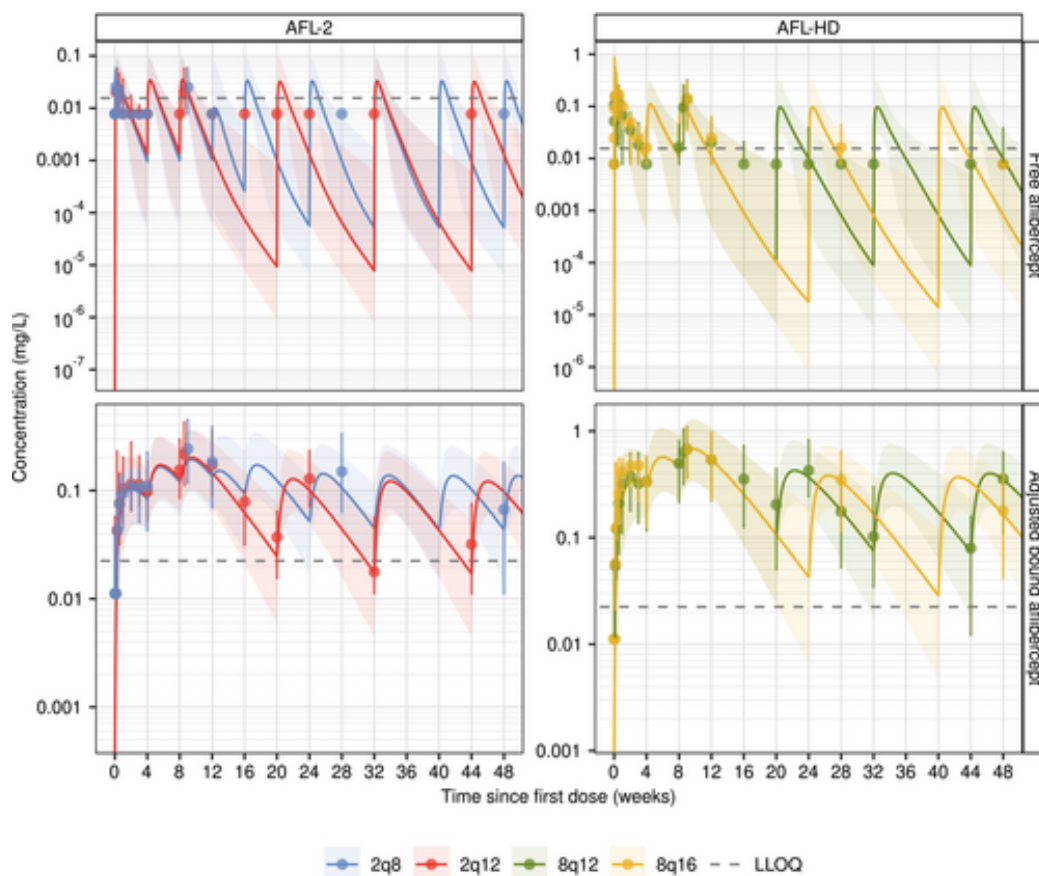


Figure 3: Comparison of median and 90% prediction interval of observed and model-predicted concentrations of free and adjusted bound aflibercept in plasma for combined nAM

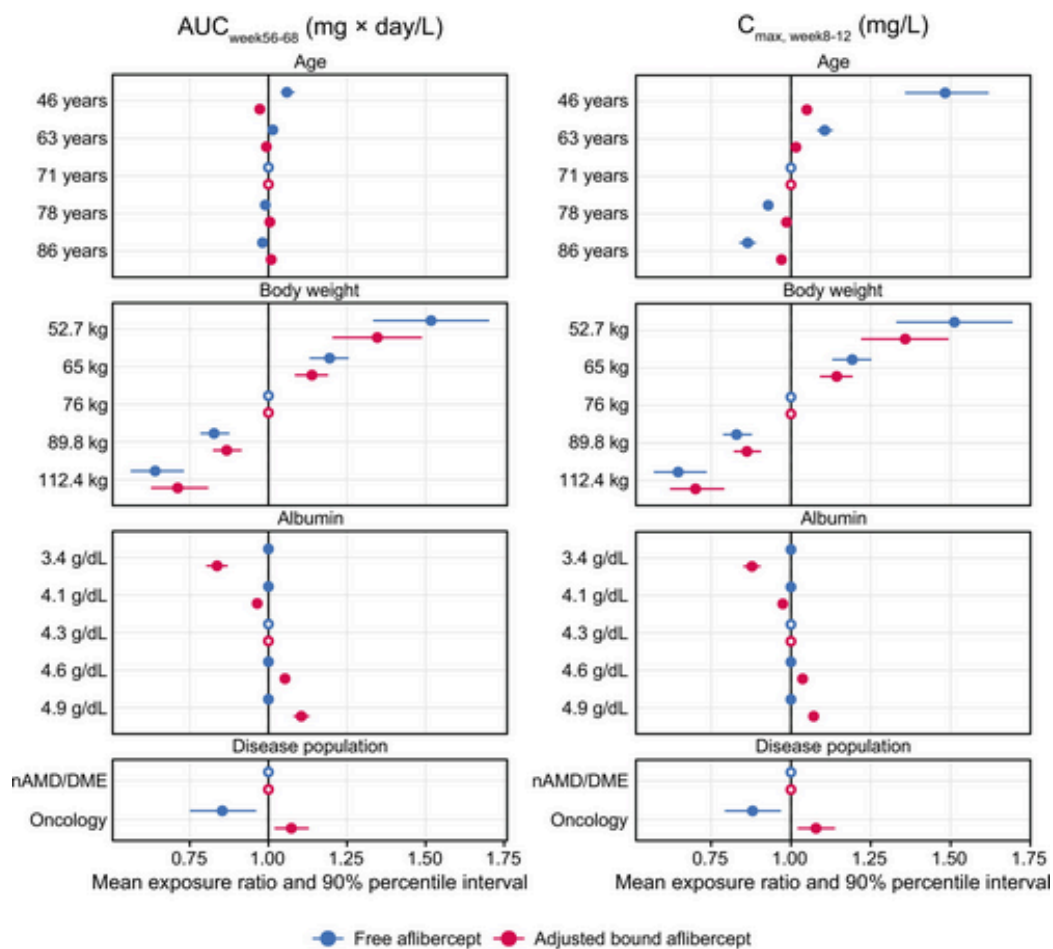


Figure 4: Typical effects of model covariates on systemic exposure metrics of free and adjusted bound aflibercept assuming 8q12 dosing in prototypical patients. The refere